

Business Insights

There is Moderate Air Quality Risk because the Average PM2.5 rests at 19.50, with "Moderate" conditions consistently present in observations, this posing ongoing health concerns for sensitive populations.

Dashboard Evidence

- Average PM2.5 = 19.50
- The Moderate EPA Air Quality category has the largest share in the treemap.

The "Carbon Monoxide Levels by Region" chart indicates that cities such as Abu Dhabi, Abuja, Accra, Addis Ababa, and Aguarray record similar high carbon monoxide values. This suggests that the urban centers experience consistently elevated carbon monoxide levels that are likely due to transport emissions, industrial activities, and population density.

Dashboard Evidence

- Carbon Monoxide by Region visual shows nearly maximum values across several cities.

The dashboard shows an average temperature of 20.21°C, while the yearly temperature trend remains almost flat from 2024 to 2026. This shows that average temperature has not changed widely during the period, showing that short-term climate conditions have remained relatively consistent.

Dashboard Evidence

- Average Temperature = 20.21°C
- Temperature by Year line chart shows minimal variation.

The "Air Quality Risk by Moon Phase" chart shows differences in the number of observations across the moon phases; the Waxing Gibbous phase being the largest proportion (approximately 23%). This suggests that environmental conditions influencing air quality may fluctuate over time rather than occurring uniformly.

Dashboard Evidence

- Waxing Gibbous accounts for approximately 23% of recorded observations.

The yearly pressure chart shows that total atmospheric pressure remains relatively constant throughout the reporting period. If combined with stable temperatures, this suggests that fluctuations in air quality are more likely driven by pollutant emissions than by large-scale atmospheric pressure changes.

Dashboard Evidence

- Pressure by Year chart remains relatively flat.
- Temperature trends also remain stable.

Recommendations

Implementation of stricter policies for emission control in the regions with high carbon monoxide levels through vehicle emission inspections and cleaner public transport would improve the air quality significantly.

With moderate air quality being dominant, the assigned agencies should strengthen public awareness that would encourage the vulnerable groups to limit their exposure during the period when the air quality is poor.

Expansion of the air quality monitoring stations in densely populated cities to identify hotspots in time, thus supporting better policy decisions.

Conducting research into the relationship between the environmental conditions and the air quality observed during the different moon phases. With understanding of the patterns, pollution forecasting may improve.

Monitoring of temperature and atmospheric pressure by the policy makers although they are stable should be conducted. This is to detect early signs of climate change and implement strategies to adapt.

Storytelling and Explanation

The dashboard presents a picture of generally stable climate conditions and persistent air quality concerns. Average temperature and atmospheric pressure have remained relatively constant throughout the reporting period, showing that there have been no major short-term climatic shifts. Air pollution indicators tell a different story.

The EPA Air Quality Index shows that Moderate air quality conditions dominate, while several cities consistently experience high carbon monoxide concentrations. This suggests that human activities, particularly transportation and urban development, are likely contributing more to environmental risk than changes in climate variables.

Additionally, the variations in air quality across different moon phases indicate that environmental conditions fluctuate over time and may be influenced by broad atmospheric or seasonal patterns. Overall, the dashboard shows the importance of prioritizing pollution reduction strategies, expanding environmental monitoring, and maintaining long-term climate surveillance to protect both public health and environmental sustainability.